

1. Simplify the Boolean function

$$F(A, B, C, D) = \prod M(3, 5, 7, 11, 13, 15)$$

using a four-variable K-map. Then draw the NOR circuit for your simplified expression, using the method we went over in class.

2. Simplify the Boolean function

$$F(x, y, z) = \sum m(0, 1, 2, 5, 7)$$

using a three-variable K-map. Write the minimum sum-of-products expression.

3. A three-variable function is given as

$$F(A, B, C) = AB + \overline{A}\overline{C} + A\overline{B}C.$$

Plot it on a three-variable K-map, read off its minterms, and find the minimum sum-of-products expression.

4. Simplify the Boolean function

$$F(w, x, y, z) = \sum m(1, 3, 5, 7, 8, 9, 12, 13, 14, 15)$$

using a four-variable K-map. Write the minimum sum-of-products expression.

5. Consider the Boolean function

$$F(A, B, C, D) = \sum m(0, 1, 2, 5, 6, 7, 8, 9, 10, 13, 14).$$

List all prime implicants, state which are essential, and write one valid minimum sum-of-products expression.

6. For the Boolean function

$$F(A, B, C, D) = \sum m(0, 1, 4, 5, 6, 7, 8, 10, 12, 14),$$

use a four-variable K-map to find (a) the minimum sum-of-products expression and (b) the minimum product-of-sums expression.

7. A three-variable function is given in product-of-maxterms form,

$$F(x, y, z) = \prod M(1, 3, 5, 6).$$

Plot it on a K-map and find the minimum product-of-sums expression.

8. Simplify the incompletely specified function

$$F(w, x, y, z) = \sum m(0, 4, 6, 12, 14), \quad d(w, x, y, z) = (2, 10)$$

to a minimum sum-of-products expression, using the don't-care conditions where helpful.

9. A circuit reads a binary-coded-decimal (BCD) digit $wxyz$ (valid inputs 0 through 9 only) and outputs $F = 1$ when the digit is 4, 5, 6, 7, 8, or 9. The input combinations 10 through 15 never occur and are treated as don't-cares:

$$F(w, x, y, z) = \sum m(4, 5, 6, 7, 8, 9), \quad d(w, x, y, z) = (10, 11, 12, 13, 14, 15).$$

Find the minimum sum-of-products expression.

10. Simplify the Boolean function

$$F(A, B, C, D) = \sum m(0, 4, 5, 6, 8, 12, 13)$$

using a four-variable K-map. Then draw the NAND circuit for your simplified expression, using the method we went over in class.

11. Simplify the Boolean function

$$F(A, B, C, D) = \prod M(0, 2, 5, 7, 8, 10)$$

using a four-variable K-map. Then draw the NOR circuit for your simplified expression, using the method we went over in class.

12. Answer each of the following.

(a) Simplify $x \oplus x \oplus y \oplus 0$ to a minimum expression.

(b) Using the definition $x \oplus y = x\bar{y} + \bar{x}y$, prove the identity

$$\overline{x \oplus y} = x \oplus \bar{y}.$$

You may use a truth table or Boolean algebra.

(c) Prove that XOR cancels itself, that is, show $x \oplus (x \oplus y) = y$ for all values of x and y .

(d) Express the three-variable function $A \oplus B \oplus C$ as a sum of minterms, $\sum m(\dots)$, and explain why these particular minterms appear (think about the odd function).

(e) We want to check whether two 5-bit numbers $A = A_4A_3A_2A_1A_0$ and $B = B_4B_3B_2B_1B_0$ are equal. Using XNOR gates, write the Boolean expression for the output *Equal*, then use it to determine whether $A = 11010$ and $B = 11000$ are equal.

(f) A 5-bit message $vwx yz = 10110$ is sent using odd parity. Find the parity bit P and write the full transmitted word $vwx yz P$.